# Mironshoh Inomjonov

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## Experience

**Research Engineer** *Apr. 2025 - Present*

*Tilmoch.ai - Tashkent, Uzbekistan*

- Conducting research on LLM alignment, post-training methods for MT models and building large scale data pipelines.
- Implemented the Contrastive Preference Optimization (CPO) fine-tuning approach, resulting in an increase in 5.2 COMET score over the baseline model.
- Developed an end-to-end parallel data mining pipeline for MT: automated sentence alignment of literary translations via LaBSE cosine similarity. Deployed on GCP (Compute Engine + Cloud Storage).

**ML Engineer** *Dec. 2024 - Apr. 2025*

*Vention - Tashkent, Uzbekistan*

- Developed real-time computer vision systems for object and action detection in surgical environments using YOLO, MobileNet, and ViT architectures, achieving high accuracy on 15 FPS video streams.
- Designed a classification algorithm using Laplacian variance to distinguish surgical smoke from fogged camera lens conditions for reliable real-time visual quality assessment during procedures.
- Developed end-to-end Uzbek STT and TTS pipelines by fine-tuning Whisper-Medium and Microsoft T5-TTS models on a single NVIDIA T4 GPU, achieving a WER of 20% with only 10 hours of speech data, demonstrating strong performance in a low-resource language setting.

**ML Engineer** *Apr. 2023 - Dec. 2024*

*KPI.com - London (Remote)*

- Implemented a chatbot using RAG with caching, reducing mean response time from 5 seconds to 2 seconds.
- Developed an image-based invoice parsing system by fine-tuning the Donut model and deploying it on AWS EC2 for scalable production use.
- Engineered back-end system for every ML pipeline using FastAPI.

## Education

**Integrated MSc/PhD, Computer Science and Engineering** *Starting from: Sept. 2026*

*Sungkyunkwan University (SKKU) - Suwon, South Korea*

- Member of InfoLab. Research areas: multimodal learning, explainable AI.

**Bachelor of Science, Software Engineering** *2021 - 2025*

*New Uzbekistan University - Tashkent, Uzbekistan*

- Recipient of the Presidential Scholarship (full undergraduate tuition).
- Relevant coursework: Machine Learning, Artificial Intelligence, Linear Algebra, Probability & Statistics.

## Publications&Projects

Paper: Filling the Gap for Uzbek: Creating Translation Resources for Southern Uzbek*EMNLP 2025 (WMT-2025)*

- Paper accepted to EMNLP 2025 (WMT-2025) conference.
- Link to paper: https://www2.statmt.org/wmt25/pdf/2025.wmt-1.83.pdf

Neurograd: Automatic differentiation Deep Learning framework inspired by Karpathy's micrograd.

- Run pip install neurograd-vec to install framework.
- Implements reverse-mode automatic differentiation over a dynamically built DAG, extending scalar micrograd-style autodiff to vectors. Includes a minimal neural-network library with a PyTorch-like API